ISPRS Journal of Photogrammetry and Remote Sensing 147 (2019) 267–282

Contents lists available at ScienceDirect

ISPRS Journal of Photogrammetry and Remote Sensing

journal homepage: www.elsevier.com/locate/isprsjprs

Evaluating the capability of the Sentinel 2 data for soil organic carbon prediction in croplands

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ARTICLE INFO

Keywords:
Sentinel-2
Soil organic carbon mapping
Hyperspectral data
Multispectral data
SNR

ABSTRACT

The short revisit time of the Sentinel-2 (S2) constellation entails a large availability of remote sensing data, but S2 data have been rarely used to predict soil organic carbon (SOC) content. Thus, this study aims at comparing the capability of multispectral S2 and airborne hyperspectral remote sensing data for SOC prediction, and at the same time, we investigated the importance of spectral and spatial resolution through the signal-to-noise ratio (SNR), the variable importance in the prediction (VIP) models and the spatial variability of the SOC maps at field and regional scales. We tested the capability of the S2 data to predict SOC in croplands with quite different soil types and parent materials in Germany, Luxembourg and Belgium, using multivariate statistics and local ground calibration with soil samples. We split the calibration dataset into sub-regions according to soil maps and built a multivariate regression model within each sub-region. The prediction accuracy obtained by S2 data is generally slightly lower than that retrieved by airborne hyperspectral data. The ratio of performance to deviation (RPD) is higher than 2 in Luxembourg (2.6) and German (2.2) site, while it is 1.1 in the Belgian area. After the spectral resampling of the airborne data according to S2 band, the prediction accuracy did not change for four out of five of the sub-regions. The variable importance values obtained by S2 data showed the same trend as the airborne VIP values, while the importance of SWIR bands decreased using airborne data resampled according the S2 bands. These differences of VIP values can be explained by the loss of spectral resolution as compared to APEX data and the strong difference in terms of SNR between the SWIR region and other spectral regions. The investigation on the spatial variability of the SOC maps derived by S2 data has shown that the spatial resolution of S2 is adequate to describe SOC variability both within field and at regional scale.

1. Introduction

Soil organic carbon (SOC) is a key property for soil quality and food production, as recognized by the European Union who considered the decline of SOC in European soils as one of the main threats for soil degradation (CEC, 2006). Furthermore, an increasing demand for monitoring carbon levels particularly in croplands is needed, as the depleted C levels provide an opportunity for carbon sequestration through adequate management practices (Lal, 2004). Unfortunately, the costs related to field sampling and laboratory analysis restricts the monitoring of soil properties at large scale (Conant et al., 2011). Spectroscopy from a remote sensing platform has been widely recognized as a rapid and effective way to quantify the amount of SOC in bare soils. At least five satellites equipped with hyperspectral imagers in the optical domain will be launched in the near future: the PRecursore IperSpettrale della Missione Applicativa (PRISMA; Pignatti et al., 2015) of the Italian Space Agency (ASI), Environmental Mapping and Analysis Program (EnMAP; Guanter et al., 2015) of the German Aerospace Center (DLR), the Hyperspectral Imager Suite (HISUI; Tanii et al., 2012) of the Japan aerospace exploration agency (JAXA), the U.S. NASA Hyperspectral Infrared Imager (HyspIRI; Houborg et al., 2012), the China Commercial Remote-sensing Satellite System (CCRSS) and the Israeli Hyperspectral imager (SHALOM; Staenz et al., 2013). These satellites will open the way for quantitative global mapping and

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https://doi.org/10.1016/j.isprsjprs.2018.11.026
Received 9 August 2018; Received in revised form 16 October 2018; Accepted 29 November 2018
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monitoring of SOC content from space in exposed soils. Nevertheless, high quality multispectral sensors such as Sentinel-2 (S2) have been launched which capacity to quantify SOC content may match the one of upcoming hyperspectral spaceborne sensors.

Currently, after the deactivation of Hyperion in 2017, there are no active hyperspectral satellite imagers covering visible (VIS), near infra-red (NIR) and short wave infra-red (SWIR) regions, consequently it is difficult to demonstrate the advantage of using satellite hyperspectral sensors compared to multispectral ones. The comparison between models relying on remote sensing data acquired using different spatial and spectral resolution is hardly addressed. Castaldi et al. (2016) evaluated the potential of some multispectral (ALI, Sentinel-2 and Landsat 8) and hyperspectral (Hyperion, EnMAP, PRISMA and HyspIRI) imagers to estimate soil texture and soil organic carbon (SOC). They simulated remote sensing spectra (adding noise and atmospheric disturbance) from laboratory spectra and their results suggest an improvement of soil variable prediction using the forthcoming hyperspectral sensors compared to the accuracy that can be obtained by multispectral or Hyperion sensors. Castaldi et al. (2014) compared Hyperion (hyperspectral) and Advanced Land Imager (ALI; multispectral) satellite data acquired at the same time, over the same area and both at 30-m spatial resolution. In this case, the spectral resolution and range degradation from hyperspectral to multispectral entailed a slight increase in the prediction error for clay and sand, while the differences between the sensor performances were more evident for soil organic matter. These weak differences were partly due to the low signal-to-noise ratio (SNR) of the Hyperion in the SWIR region, especially between 1900 and 2500 nm (Castaldi et al., 2016; Folkman et al., 2001), where the most important clay features are located. Moreover, the ALI sensor had three bands in the SWIR region (1250, 1650 and 2100 nm), with a higher SNR than Hyperion at the same wavelengths. Other available multispectral satellites, such as the Satellite Pour l'Observation de la Terre (SPOT) series, WorldWiew-4 (DigitalGlobe) or Pléiades (CNES) provide high spatial resolution images, but they have no bands in the SWIR region.

The S2 sensors acquire thirteen bands, which cover the entire VNIR-SWIR spectral range; the three narrow VIS bands (GSD: 10 m) and the four bands in the NIR region make the S2 appropriate for surveys and investigations over vegetated areas (e.g. Vrieling et al., 2018). Although the two bands in the SWIR are quite broad, their presence in combination with the good spatial resolution in the VIS (10 m) and NIR-SWIR (20 m) regions is promising for geological applications (van der Meer et al., 2014) and soil variable mapping (Castaldi et al., 2016). The most important spectral features for SOC prediction are located in the VIS region at 450, 590 and 664 nm (Ben-Dor et al., 1997; Nocita et al., 2015; Castaldi et al., 2018a), and very close to the S2 bands in this spectral range (490, 560 and 665 nm). Other useful features for SOC prediction, related to specific chemical bonds, can be detected in the SWIR region between 1600 and 1900 nm and around 2100 and 2300 nm. The SWIR spectral features overlap with the two S2 broad bands centred at 1610 and 2190 nm. Consequently, the good match between SOC spectral features and S2 bands could be exploited for quantitative SOC prediction and mapping. However, as far as we are aware, very few studies dealt with actual S2 data for soil property prediction (e.g. Gholizadeh et al., 2018). Some authors used simulated data to investigate the iron absorption features (van der Werff and van der Meer, 2015), to estimate soil organic matter (Rosero-Vlasova et al., 2017) or soil texture and SOC (Castaldi et al., 2016). Recently some studies focused on soil moisture mapping using actual S2 data (Sadeghi et al., 2017) in synergy with Sentinel-1 data (El Hajj et al., 2017; Gao et al., 2017).

In addition to spectral characteristics of the sensors (spectral range, bandwidth, SNR, etc.), the spatial resolution also affects the prediction accuracy of soil properties derived from remote sensing data. Gomez et al. (2015) tested the sensitivity of clay content prediction to degradation of spatial resolution, simulating hyperspectral satellite data from AISA-DUAL airborne images covering the 400–2450 nm spectral domain. They found that satisfactory results for clay prediction could be obtained with a Ground Sampling Distance (GSD) from 5 to 30 m. However, the influence of spatial resolution increased when the investigated soil property has a short-scale spatial variability. Previously, Casa et al. (2013) tested airborne MIVIS data (GSD of 4.8 m) for soil texture prediction; the results achieved by the airborne sensor were not consistently different from those obtained with the same soil data from CHRIS-PROBA data with a GSD of 17 m and a similar spectral range. Similarly, Chabrillat et al. (2014) and Steinberg et al. (2016) tested the capability of simulated EnMAP spaceborne hyperspectral imagery (30 m) compared to Airborne Hyperspectral Scanner (AHS) imagery (2.6 m) for among others SOC quantitative prediction. Steinberg et al. (2016) demonstrated that, despite a slight decrease in prediction accuracy at the spaceborne scale, the decrease in SOC prediction accuracy was still reasonable. On the other hand, the lowering of the spectral resolution can affect the prediction accuracy derived from satellite sensors having a coarse spatial resolution. The GSD of the SWIR S2 bands (20 m) is a considerable improvement compared to that provided by the SWIR Landsat 8 bands (30 m).

The critical factors to retrieve soil and geological properties by hyperspectral sensors are the SNR and the spatial resolution (Transon et al., 2018), provided that spectral coverage and resolution are adequate. Consequently, the recent findings suggest paying more attention to multispectral satellites for soil property prediction. In particular, the short revisit time of the S2 constellation entails a large amount of data freely available for the scientific community.

In this regard, we tested the capability of actual S2 data to predict SOC in croplands areas. We selected three regions in Germany, Luxembourg and Belgium for which hyperspectral airborne data were available. We compared the S2 results with those obtained by hyperspectral airborne data both using full spectrum data or spectra resampled according to S2 bands. Thus, this study aims to compare the capability of multispectral and hyperspectral remote data for SOC prediction, and at the same time, we investigated the importance of spectral and spatial resolution through the SNR, the variable importance in the prediction models as well as the spatial variability of the SOC maps at field and regional scale.

2. Materials and methods

2.1. Study area

We chose three agricultural areas in three different European countries: Germany, Belgium and Luxembourg (Fig. 1). These areas were chosen because of the availability of hyperspectral airborne and ground-truth data. Moreover, they have different soil characteristics and a considerable variability of SOC content.

The German study area consists of a 2000 km^2 square located in the lowlands of north-eastern Germany around the town of Demmin (53°52′ N; 13°13′E). This test site belongs to the German observatory network TERENO (Zacharias et al., 2011) that provides long-term statistical series of system variables for the analysis and prognosis of Global Change consequences within the 15 years observation term of TERENO. The Demmin area is an agricultural ecosystem where a three years rotation of winter cereals (about 60%) and root crops prevails. Grassland and pasture cover about 25% of the area. Fields are large and are bare during a short period before and after the seeding in spring for maize, sugar beet and potato and in autumn for the winter cereals or cover crops. The topography is rolling in the north and hilly to undulating in the south with an altitudinal variation of 120 m. Considerable differences in the parent material and relief caused a high spatial variability of soil types in some parts. The flat and slightly undulating plains are characterized by sand rich regions and extensive areas with glacial till. The sandy regions are dominated by Cambisols, Luvisols and Albeluvisols. Luvisols, Albeluvisols, Stagnosols (IUSS Working Group

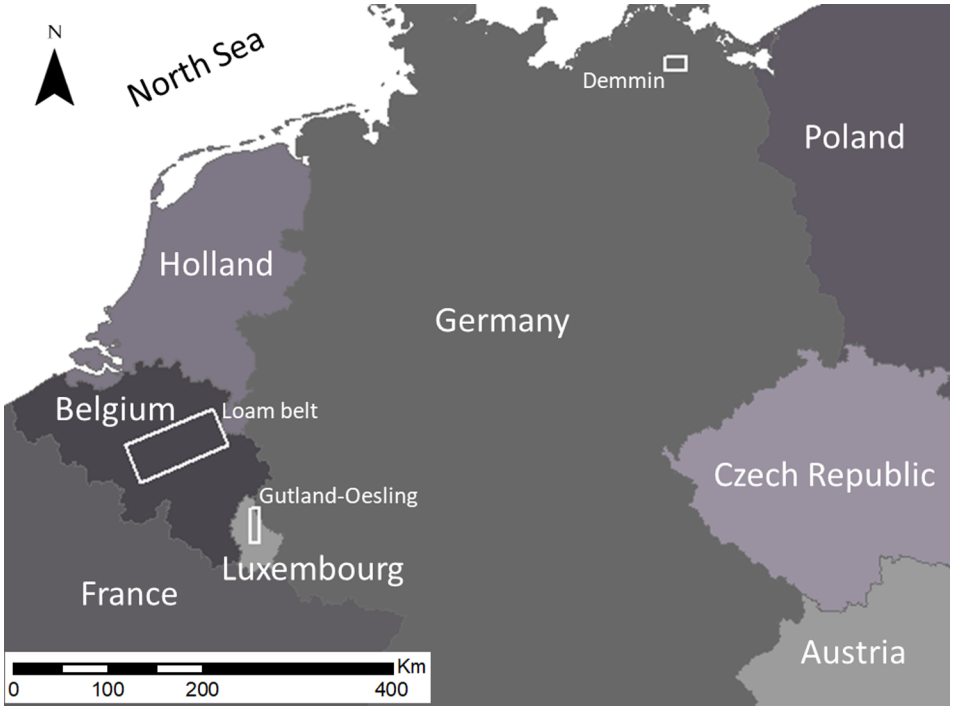


Fig. 1. Location of the three study areas in Germany, Belgium and Luxembourg.

WRB, 2015) evolved on glacial tills. Soils on glacial till in hilly terrain are often truncated and colluvial sediments have accumulated at the bottom of the slopes. The floodplains which are mainly used as grassland are characterized by organic and shallow peat soils. The long-term mean temperature of the area is about 8 °C with an annual precipitation from 500 to 600 mm increasing from south-east to north-west (Gerighausen et al., 2012).

The Belgian study area is located in the loam belt in the northern part of Wallonia. The region is dominated by cropland on niveo-eolian loess with a rolling topography. The uniformity of the substrate entails that most of the area is characterized by silt loam and well drained soils. The area consists of a 9.7 km-wide and 44 km-long strip (NW corner: 50.60N, 4.65E; SE corner: 50.75N, 5.23E). The climate in this region is temperate oceanic with mean temperatures between 2.3 °C (January) and 17.8 °C (July) and a mean annual precipitation of 790 mm (Castaldi et al., 2018b).

The Luxembourg area is located in two geographical regions: Gutland in the centre and Oesling in the northern part of the Grand Duchy of Luxembourg. The investigated area cover a narrow North–South strip covering an area of 6 × 34 km (NW corner: 50.04N, 6.08E; SE corner: 49.73N, 6.18E). Limestone, sandstone, secondary marls and dolomites characterize the Gutland region. The soils are silt loam, clay and clay-loam and they become sandier in the southern part of the region where the substrate mainly consists of calcareous sandstone. The soils of the Oesling region are mainly shallow stony loam soils having a structural B horizon. These soils originated from Devonian slate, quartzite and sandstone substrates on a sub-horizontal plateau having a mean altitude of 450 m. The climate in Luxembourg is temperate oceanic. The mean annual precipitation is approximately 900 mm and the mean temperatures ranges from 0.7 °C in January to 17 °C in July (Stevens et al., 2012).

The soils in the Belgian loam belt are quite homogenous except for some variation in drainage. Consequently, for the Belgian dataset it was not necessary to split the soil samples into subsets. While, both in Germany and in Luxemburg, we split the samples into two subsets according to the main soil associations derived from the broad scale maps: the Soil Map of Great-Duchy of Luxembourg (1:100,000) (LSP, 1969) and the Soil Map of the Federal Republic of Germany (1: 1,000,000) (BGR, 2007). The German soil map distinguishes two sub regions: one with clay soil (Demmin_C) and another one characterized by a sandy layer (Demmin_S). The Luxembourg samples were collected within two regions quite different in terms of geological substrate: Oesling and Gutland.

2.2. Remote sensing images

2.2.1. Sentinel-2 images

Currently, the S2 constellation consists of two identical sensors acquiring data between 443 and 2190 nm at five days of revisit time. Other two S2 sensors will be launched in the next years. The thirteen S2 bands have different spatial resolution: B2 (490 nm), B3 (560 nm), B4 (665 nm) and B8 (842 nm) have a spatial resolution of 10 m, while the 20 m spatial resolution bands are B5 (705 nm), B6 (740 nm), B7 (783 nm), B8a (865 nm), B11 (1610 nm) and B12 (2190 nm). The other three bands, the atmosphere correction channels, have 60 m spatial resolution (B1, B9 and B10).

Three cloud-free Sentinel-2 images were downloaded from the Copernicus open access hub (Table 1), with similar crop exposure conditions than during the airborne acquisitions. The images were provided as Level-1C product, thus in Top of Atmosphere (TOA) reflectance. In order to obtain a Level-2A production, i.e. Bottom of Atmosphere (BOA) reflectance, the images were atmospherically corrected using the Sen2Cor processor (v. 2.5.5) (Mueller-Wilm et al., 2018), a plugin incorporated into the Sentinel Application Platform (SNAP) software. The standard Sen2Cor rural aerosol mode was selected for all images. The atmospherically corrected images were

Table 1
Acquisition times of the Sentinel-2 and airborne images (see Fig. 1 for the study areas).

Study area	Acquisition date Sentinel-2	Acquisition date airborne
Gutland-Oesling (Luxembourg)	2016/05/08	2015/03/20
Demmin (Germany)	2017/08/30	2015/10/01
Loam belt (Belgium)	2017/08/29	2015/07/01

spatially resampled at 10 m to maximise the level of detail of the S2 data. We selected nine S2 bands: B2 (490 nm), B3 (560 nm), B4 (665 nm), B5 (705 nm), B6 (740 nm), B7 (783 nm), B8 (842 nm), B11 (1610 nm) and B12 (2190 nm). In order to correct the geometrical shift of the S2 images and thus obtain the overlap between S2 and airborne image, the S2 images were geometrically registered using the airborne data as reference images.

2.2.2. Airborne images

The Airborne Prism Experiment (APEX) sensor acquired the Belgium and Luxembourg imagery mounted on a non-pressurized Dornier 228 aircraft of the DLR German Aerospace Center. The APEX is a hyperspectral dispersive push broom airborne imaging spectrometer recording data in 313 bands (in binned mode) between 400 nm and 2500 nm. The data were recorded on 1 July 2015 in Belgium (five flight lines at flying height 3900 m) between 15:00 and 16:00 and on 20 March 2015 in Luxembourg (three flight lines at flying height 4700 m) between 14:00 and 14:30, both in clear sky conditions, resulting in a sampling distance of respectively about 2.0 and 2.3 m across track.

The APEX data were pre-processed at the VITO Remote Sensing Department, using the Central Data Processing Center (CDPC) for airborne earth observation data (Biesemans et al., 2007). During a pre-campaign calibration session at the APEX Calibration Home Base (CHB) hosted at DLR Oberpfaffenhofen, Germany (Gege et al., 2009), calibration cubes were generated in order to perform the APEX radiometric and spectral calibration. Global Position System (GPS) and Inertial Measurement Unit (IMU) data were registered during the flight and were post processed with Applanix POSPac MMS software thereby using GPS base stations information. The ortho-rectification process uses these post-processed position and orientation data together with boresight correction data, the detailed sensor model and the digital elevation model (DEM) of the Shuttle Radar Topography Mission (SRTM). Finally, the images were spatially resampled to an output resolution of 2.5 m and projected to the geographic coordinate system (WGS84). The atmospheric correction of the APEX data was performed using the MODTRAN4 radiative transfer code and reflectance retrieval algorithms given in de Haan et al. (1991) and de Haan and Kokke (1996). As a result, the pre-processed image data were provided as geo-referenced at-surface-reflectance.

The SNR of the APEX (Schaepman et al., 2015) imagery was computed in the quality assessment of the APEX processing and archiving facility (Hueni et al., 2009). The APEX SNR model was developed in the framework of the MetEOC-1 project (NPL) (Hueni et al., 2013). It is based on integrating sphere measurements taken in the APEX Calibration Home Base (Gege et al., 2009) at various radiance levels to cover as much as possible of the dynamic range expected under flight conditions. Measurement data were stored and processed to radiance level in the APEX Calibration Information System (Hueni et al., 2013), disregarding any saturated frames. The SNR per at-sensor radiance level was computed via the mean radiance divided by the standard deviation of 20–30 frames. The SNR model was then parameterized by fitting first and third order polynomials to all spectral bands, building a transfer function between at-sensor radiance and SNR. It must be noted that the current SNR model represents the mean SNR of all across-track pixels, hence, the SNR numbers provided by the model should be treated as approximations.

The German Demmin airborne images were acquired on 1 October 2015 with the HySpex system of the Helmholtz Center Potsdam GFZ, mounted on the Cessna 207T aircraft of the Free University of Berlin during a flight campaign in preparation for EnMAP. The NEO HySpex system consists of two push-broom hyperspectral cameras (VNIR-1600 operating over the 400–1000 nm and SWIR 320 m-e operating over 1000–2500 nm range) with a total of 416 wavebands and a spectral resolution of 3.7 nm (VNIR-1600) and 6.0 nm (SWIR-320 m-e). From a mean altitude of 2500 m, the original ground sampling distance (GSD) of the image was 1.9 m for the VNIR spectrometer and 4.0 m for the SWIR-320 m-e camera, resampled to 4 m after data pre-processing. The pre-processing of the HySpex data to orthorectified reflectance was performed with the GFZ in-house processing chain HyPrepAir. First, geometric processing was performed, including co-registration and adaptation of the SWIR sensor to the VNIR, based on the procedures from Brell et al. (2016). Subsequently, atmospheric correction of the HySpex VNIR-SWIR data cube was applied with the ATCOR-4 software (Richter and Schläpfer, 2016). Then, a mosaicking of the single flight lines keeping the original data values was realized. Further, an Empirical Line Calibration (ELC) was performed using ground measurements obtained simultaneous to flight acquisition to remove atmospheric attenuation and spectral artefacts. Unfortunately, only very basic laboratory SNR characterization is available for the Hyspex sensors, which is not sufficiently transferable to the Demmin flight campaign. Additionally, the study area Demmin does not contain large and homogeneous surfaces to approximate the effective SNR in a reliable manner. Therefore, no quantitative statement regarding the SNR is made for the Demmin flight campaign. However, the flight campaign was realized in early October and due to non-optimal illumination conditions and relatively dark soil surfaces, the SNR cannot be considered as optimal and the readout noise is getting relevant.

2.3. Soil samples

The soil sampling campaigns for ground-truthing were carried out in 2015 in Belgium (170 samples) and in 2016 in Luxembourg (194 samples), and in 2017 in Germany (231 samples). We randomly collected at least one sample for each selected field excluding areas close to field boundaries to avoid border effects. Each sample consisted of five sub-samples taken with a gouge auger within an area of 5 m radius and collected from 0 to 10 cm depth. Soil samples were air dried and passed through a 2 mm sieve in the laboratory, and then total carbon was measured by dry combustion using a CN analyser (VarioMax, Elementar Gmbh, Hanau, Germany). The carbonate content was determined by measuring the pressure of CO_2 emitted after the addition of hydrochloric acid (Sherrod et al., 2002) and this value was subtracted from the total carbon to obtain SOC content. We selected samples collected in soils that were bare at S2 or airborne acquisition date (Fig. 2). Soil conditions at sampling points (bare or crop) were assessed by visual inspection for the airborne data and computing the NDVI for the S2 images. Within the three areas (Germany, Luxembourg and Belgium), the soil samples were further split into subsets according to soil associations derived from available broad scale maps (Fig. 2).

The SNR values for the two APEX flight campaigns were extracted at soil sampling points collected in Luxembourg and Belgium. Consequently, we computed the average SNR values for each wavelength, thus obtaining the average SNR spectrum for each flight campaign (Luxembourg and Belgium). The SNR values at sampling points were also resampled according the S2 bands taking into account the enhancement of the SNR (Seidel et al., 2018) due to the broader S2 compared to the APEX bands.

2.4. Soil organic carbon predictive models

For every dataset, the SOC content measured in each soil sample in the laboratory was paired to the spectrum extracted from the S2 and airborne sensor at the sampling point. These data were used to build a SOC predictive model (Fig. 3).

The predictive models concern three types of spectral data:

- Multispectral S2 data
- Hyperspectral airborne data (APEX or HySpex)
- Airborne data (APEX or HySpex) resampled according S2 bands using the central wavelengths and the full width half maximum (FWHM; Pahlevan et al., 2017). The spatial resolution remains the same of the airborne data.

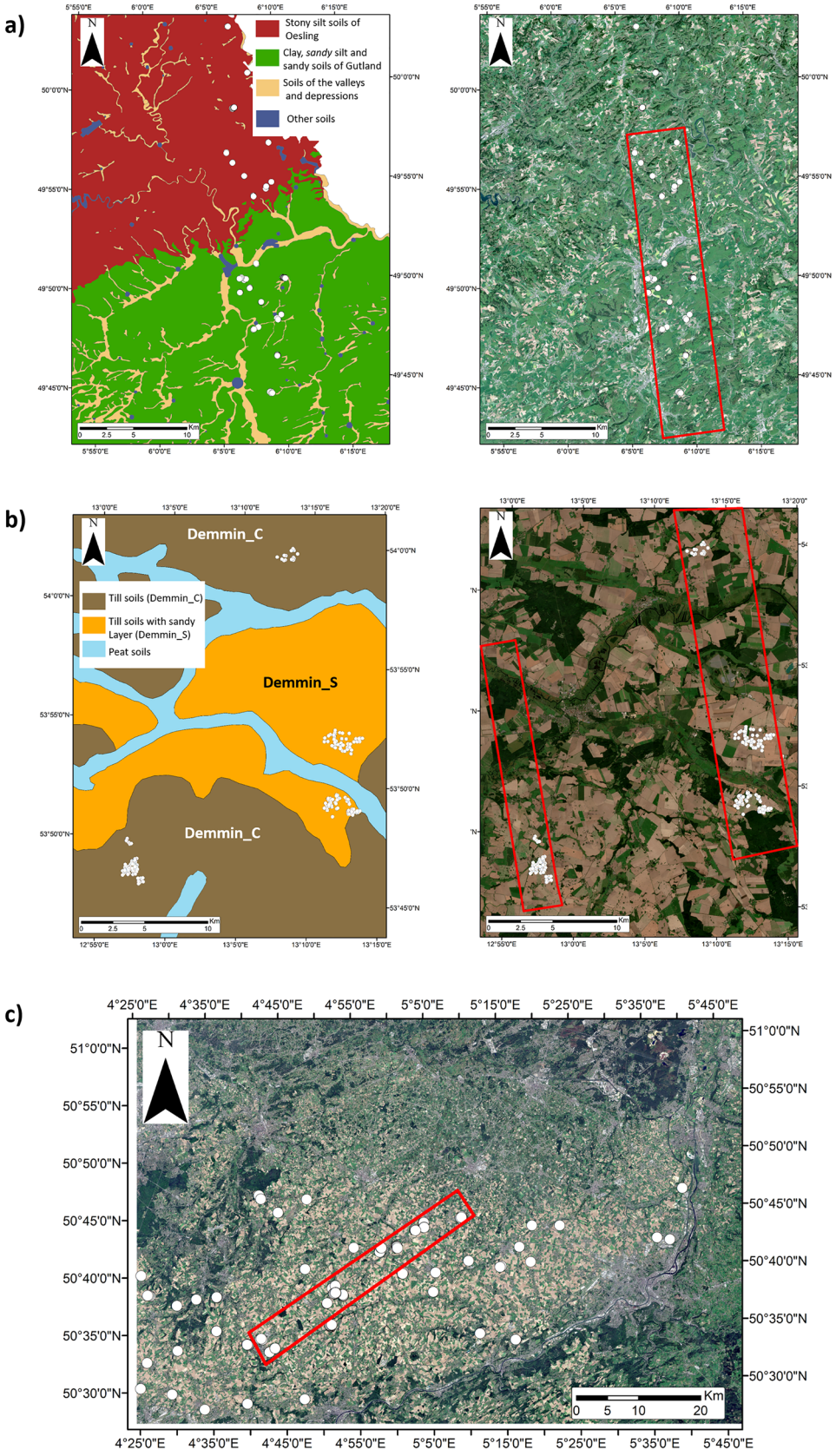


Fig. 2. Soil types maps and RGB (red: 665 nm; green: 560 nm; blue: 490 nm) and satellite images acquired by Sentinel-2 sensor in Luxembourg (a), Germany (b) and Belgium (c). The red shapes show the flight areas from the airborne acquisitions. The white points indicate the location of the soil sampling points. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

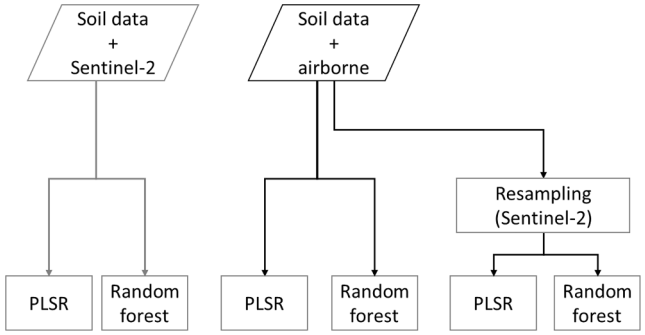


Fig. 3. Flowchart concerning the spectral data and the soil organic (SOC) prediction approaches used.

Two different multivariate models were tested for all spectral data: partial least square regression (PLSR) (Wold et al., 2001) and random forest (RF) (Breiman, 2001), and the variable importance was calculated for each regression method.

The normality of the distribution of the SOC contents was verified before testing the predictive models (skewness < 1), and for non-normal distributions a Box-Cox transformation was carried out. The full airborne spectra were pre-processed removing the water bands and applying a Savitzky–Golay smoothing filter (Savitzky and Golay, 1964) with a second-order polynomial fit and a window size of eleven data points, while no pre-treatment was applied to S2 spectra.

Partial least square regression allows building a multivariate regression model exploiting a limited number of uncorrelated components, in this case the spectral bands (Wold et al., 2001). The PLSR components were obtained taking into account both matrices of the spectral bands (independent variables) and SOC content (dependent variable). The number of optimal PLSR components was set selecting what provided the lowest root mean square error (RMSE; Eq. (1)) of 10-folds cross validation. The variance importance in projection (VIP) values were calculated to estimate the importance of each band in the PLSR model (Wold et al., 2001). The VIP value is the weighted sum of squares of the PLSR weights, calculated taking into account the amount of explained variance of each component of the model. Since the average of squared VIP values is equal to 1, independent variables (bands) showing a VIP value greater than 1 are considered significant for the prediction model (Chong and Jun, 2005).

Random forest is an ensemble learning technique that is widely used for both classification and regression (Breiman, 2001). The RF regression is a supervised learning algorithm that exploits the resampling of the training dataset to combine multiple regression trees, which make up the 'forest'. In this case, the training dataset consist of the measured SOC values (dependent variable) and the remote sensing data (satellite or airborne) extracted at sampling points (independent variables). The predicted SOC values will be obtained averaging the outputs of the individual trees. This regression method avoids issues related to overfitting, noise or irrelevant features. The importance of each band in the RF regression model was evaluated by the relative variable importance (RVI). The differences between the mean squared error (MSE) of prediction before and after permuting each predictor was calculated and then, the mean value of the differences of all trees were normalized over the standard deviation for each variable. These values were divided by their sum obtaining the RVI for each band. Variable having RVI values greater than 1/N, where N is the number of bands, were considered as being important for the RF model, since 1/N is the expected band importance in a model where all variables are equally ranked.

The prediction accuracy for the two regression techniques was evaluated by the RMSE (Eq. (1)) of 10-folds cross-validation and the ratio of performance to deviation (RPD; Eq. (2)). The method providing the highest RPD value, between RF and PLSR, was chosen for each dataset.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_o - y_p)}{n}} \quad (1)$$

$$RPD = \frac{std}{RMSE} \quad (2)$$

where y_0 are the SOC observed values, y_p the SOC predicted by the model, *std* the standard deviation of the observed SOC values and *n* the number of samples.

2.5. Soil organic carbon regional and in-field variability

Soil organic carbon maps were obtained applying the predictive models to all bare soil fields of the same sub regions. In particular, the SOC maps were used to analyze and quantify the regional and in-field variability of the SOC content exemplary in the Demmin area. The Demmin region was chosen because of its large variability in terms of SOC content. We selected an area of about 2000 km^2 in this region, in which we masked everything that was not bare cropland soil at S2 acquisition time. We considered bare soil pixels if the NDVI value was lower than 0.25. For the in-field variability investigation we have chosen a very large field (107 ha) within the Demmin_C sub-region. The SOC predicted values obtained by S2 data for each bare soil pixel were modelled by using the methods of geostatistics, in which each SOC value $z(\boldsymbol{x}_\alpha)$ at different location $\mathbf{x}_\alpha$ ($\mathbf{x}$ denotes the coordinates in two dimensions and $\alpha = 1, \ldots, N$ are the measured points) is interpreted as a particular realization of a random variable. The correlation in space is estimated by the experimental variogram $\gamma(\boldsymbol{h})$ from data as function of the distance ($\mathbf{h}$) and direction of data pairs values $[z(\mathbf{x}_\alpha), z(\mathbf{x}_\alpha + \mathbf{h})]$ and the semivariances calculated at a set of discrete distances (lags) between couples of SOC values. A theoretical function, called variogram model, is fitted to the experimental variogram to allow the analytical estimate of the variogram for any distance $\mathbf{h}$. Since a few exceptionally large values may contribute to many very large squared differences, skewed data were transformed into a Gaussian-shaped variable with zero mean and unit variance (experimental variance) (Chilés and Delfiner, 2012). All geostatistical analyses were performed using the software package ISATIS, release 2018.

3. Results

3.1. Soil datasets

The five soil datasets are quite different in terms of SOC content

Table 2
Descriptive statistics of the soil organic content and number of samples for the Sentinel-2 and airborne datasets in the three study areas.

			SOC g kg^{-1}			
Image	Study area	n° samples	min	max	mean	sd
Sentinel-2	Oesling	44	16.8	48.3	26.7	6.2
	Gutland	63	6.9	25.6	14.7	3.9
	Gutland-Oesling	107	6.9	48.3	19.6	7.8
	Demmin_S	109	6.0	38.7	12.8	4.9
	Demmin_C	71	6.0	196.4	29.4	39.6
	Demmin	180	6.0	196.4	19.3	26.3
	Loam belt	115	6.6	18.6	10.7	2.8
Airborne	Oesling	44	16.8	49.9	32.9	8.5
	Gutland	75	6.9	25.3	15.1	4.5
	Gutland-Oesling	119	6.9	49.9	21.7	10.7
	Demmin_S	149	6.0	38.7	12.5	4.4
	Demmin_C	65	7.6	196.4	25.6	39.2
	Demmin	214	6.0	196.4	16.5	22.7
	Loam belt	98	6.1	19.4	10.6	2.2

Table 3
Prediction accuracy in terms of root mean square error (RMSE) and ratio of performance to deviation (RPD) obtained by cross-validation of partial least square regression (PLSR) and random forest (RF) models using Sentinel-2 and airborne data (hyperspectral and resampled according Sentinel-2 bands).

	Sentinel-2				Airborne hyperspectral				Airborne S2 bands		
Study area	n°	Model	RMSE g kg^{-1}	RPD	n°	Model	RMSE g kg^{-1}	RPD	Model	RMSE g kg^{-1}	RPD
Oesling	44	PLSR	3.7	1.7	44	PLSR	5.0	1.7	PLSR	8.3	1.0
	44	RF	4.7	1.3	44	RF	9.3	1.0	RF	9.3	1.0
Gutland	63	PLSR	2.4	1.6	75	PLSR	3.2	1.4	PLSR	3.2	1.4
	63	RF	2.5	1.5	75	RF	3.3	1.4	RF	3.4	1.3
Gutland-Oesling	107		3.0	2.6	119		4.0	2.7		5.7	1.9
Demmin_S	109	PLSR	4.7	1.0	149	PLSR	3.8	1.2	PLSR	3.7	1.2
	109	RF	4.8	1.0	149	RF	4.0	1.1	RF	4.1	1.1
Demmin_C	71	PLSR	25.2	1.5	65	PLSR	33.7	1.1	PLSR	51.2	0.8
	71	RF	18.6	2.1	65	RF	18.4	2.1	RF	18.5	2.1
Demmin	180		12.2	2.2	214		10.6	2.1		10.7	2.1
Loam belt	115	PLSR	1.9	1.1	98	PLSR	1.7	1.3	PLSR	1.7	1.3
		RF	2.0	1.0	98	RF	2.0	1.1	RF	2.0	1.1

(Table 2). The maximum variability was detected in Demmin_C, where the SOC ranges between 6 and 196.4 g kg^{-1} (taking into account both S2 and airborne dataset), while Demmin_S showed lower SOC content. The loam belt is the most homogenous area and the standard deviation is only 2.2–2.8 g kg^{-1}. The two regions in Luxembourg are quite different: SOC ranges between 16.8 and 49.8 g kg^{-1} in Oesling and between 6.9 and 25.6 g kg^{-1} in Gutland.

3.2. Prediction accuracy

For the S2 and airborne data, the best prediction accuracies were obtained using the PLSR method, except for Demmin_C region, where the RF provided the best results (Table 3). The differences between PLSR and RF models in terms of RPD are negligible in Gutland and Demmin_S, while for the Oesling and Demmin_C dataset the gap between the two regression methods is more considerable (Table 3). Although the RMSE in loam belt is quite low (1.9 g kg^{-1}), the RPD is unsatisfactory (1.1), in this case the predicted values are all around the average measured value (Fig. 4c). Merging the two Luxembourg regions, we obtained a RMSE of 3.0 g kg^{-1} and a RPD of 2.6. The prediction accuracy is lower in Demmin, (RMSE = 12.2 g kg^{-1}; RPD = 2.2).

The prediction accuracy obtained by airborne hyperspectral data is generally slightly higher than that retrieved by S2 data. The RPD is higher than 2 in Gutland-Oesling (2.7) and Demmin (2.1) region, while it is 1.3 in the Belgian loam belt. After the spectral resampling according to S2 bands the prediction accuracy remains the same for all the sub-regions except for Oesling, where the RMSE increases from 5.30 to 8.3 g kg^{-1}. Both S2 and airborne resampled models underestimated SOC values higher than 40 g kg^{-1} in Oesling (Fig. 4a). Sentinel-2, HySpex and resampled RF models underestimated SOC content for SOC content above 100 g kg^{-1} in Demmin (Fig. 4b). This is probably due to the strongly positive skewed distribution of the SOC values for this dataset: only four samples have SOC content higher than 100 g kg^{-1}. However, the box-cox transformation of the SOC values that was carried out before the prediction models allowed drastically reducing the skewness and obtaining quite reliable predictions.

3.3. Variable importance

The VIP values for hyperspectral airborne data showed values higher than 1 in the VIS spectral region (470–560 and 660–670 nm) for loam belt, Oesling (470–560 nm) and Demmin_S (650–700 nm) (Fig. 5). The RVI values for Demmin_C are higher than 0.003 (i.e. 1/number of HySpex bands) for most of the bands within the VIS region. However, the most important bands were detected in the SWIR region for Oesling, Gutland and Demmin_S. In Demmin_S, the VIP is high for wavelengths above 2000 nm, while in the Gutland region the VIP is higher than 1 for all the bands between 960 and 2400 nm. Some high VIP values were also detected between 1100 and 1300 nm in loam belt and Oesling, while a RVI peak can be observed at 950 nm in Demmin_C.

The VIP and RVI values obtained by S2 data had the same trend as the airborne VIP values in Gutland, Demmin_S and Demmin_C regions. For Germany and Luxembourg datasets the S2 bands centered at 1610 (B11) and 2190 (B12) nm showed high VIP and RVI values. Generally, the S2 VIP values for B2 (490 nm) and B3 (560 nm) bands were low, also for loam belt and Oesling dataset where airborne VIP showed high values in this spectral region (470–670 nm). While the B4 centered at 665 nm showed a VIP value higher than 1 for Demmin_S dataset, and also the RVI in Demmin_C exceeded the threshold value (0.11).

The VIP and RVI values obtained from airborne data resampled according to S2 bands showed some differences with the values obtained from S2 data (Fig. 5). The importance of SWIR bands decreased in all region except for Demmin_S and Gutland, where no evident differences, in terms of VIPs, could be detected between S2 and APEX spectrally resampled to S2. The red (665 nm) and NIR (7 0 5) band become more important than the SWIR bands in Oesling, while in loam belt the VIP of the band centered at 490 nm was much higher than the other ones. The RVI values of resampled data showed a great importance of NIR bands (705, 740, 783 and 842 nm) in Demmin_C, and at the same time the RVI of two SWIR bands fall down, showing values far below the threshold value of 0.11.

3.4. Signal-to-noise ratio

The average SNR spectra of APEX sensor had the same trend in all the sub regions (Fig. 6). The highest SNR values were detected in the VIS region and within the following ranges: 950–1100 nm, 1150–1300 nm and 1450–1750 nm. The Oesling and Gutland SNR values are very similar (Fig. 6a and b), while in the Loam belt (Fig. 6c) the SNR is generally higher across the entire spectrum than for the Luxembourg sub-regions. The lowest SNR values were measured in the SWIR region between 1800 and 2400 nm showing a sharp decrease from 2000 nm onwards. The SNR values obtained after resampling the APEX data according to S2 bands showed high values for all the bands in the VNIR region. The SNR values decrease until the fourth band (705 nm) and then increases in the other NIR bands. The SNR values sharply decrease for the last SWIR band in all the three sub regions. A strong difference between airborne resampled and Sentinel-2 data exists in term of SNR (Fig. 7). Since the broad bands collect much more photons than narrow bands, the resampling procedure of the APEX sensor entailed a strong enhancement of the SNR.

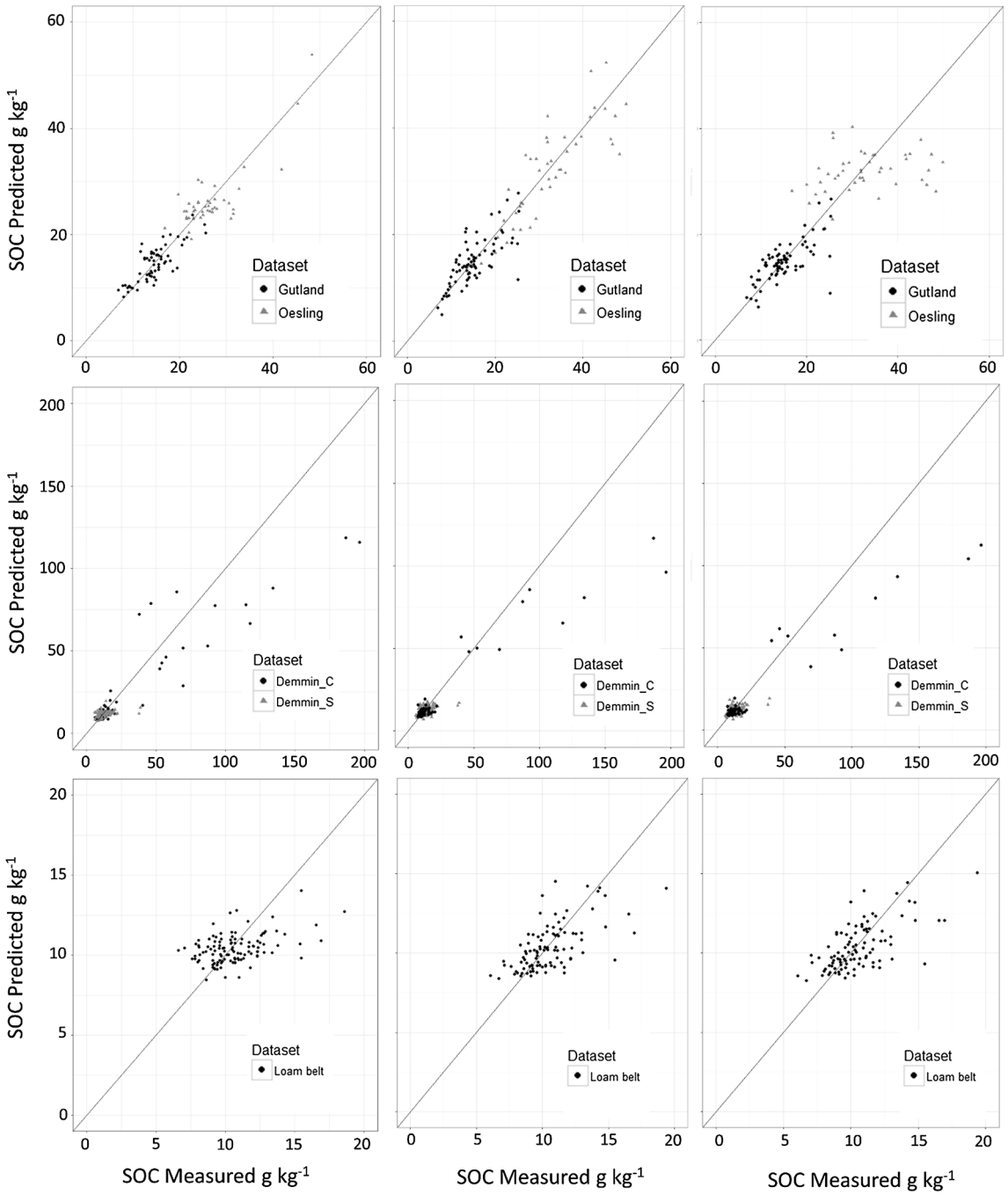


Fig. 4. Plot of measured against estimated soil organic carbon content in Luxembourg (a), German (b) and Belgian (c) datasets obtained by Sentinel-2 (left side), airborne (centre) and airborne resampled (right) data.

3.5. SOC maps variability analyses within the Demmin regions

Soil organic carbon data of the field in Demmin_C area obtained by HySpex (Fig. 8a) and actual S2 (Fig. 8b) were analysed for quantifying, modelling and interpreting their spatial dependence. The skewness coefficients of both data sets was very high and, for the subsequent analysis, SOC data were transformed into a Gaussian-shaped variable using a Gaussian anamorphosis (Wackernagel, 2003).

In the variographic analysis, no anisotropic behaviour was detected for both S2 and airborne data up to 400 m. Consequently, two isotropic nested models were fitted to the experimental variograms (Fig. 9 and Table 4).

The fitted variogram for field HySpex SOC data combines three basic structures (nested model) including a nugget effect, an exponential model at short range and a spherical model at longer range (Webster and Oliver, 2007) (Table 4). For the S2 SOC data, the fitted variogram shows no discontinuity at origin and includes two exponential models at short and longer ranges (Table 4).

In addition, the SOC data at regional scale retrieved from S2 data were analysed to explore the dependence in all the directions. The 2D variograms map showed two main direction of anisotropy along north-south (N-S) and east-west (E-W) directions. Omnidirectional and directional variograms were computed (Fig. 10) and from a visual inspection a concavity was evident in the omnidirectional (Fig. 10a) and N-S experimental variograms (Fig. 10b).

Both experimental variograms (omnidirectional and directional)

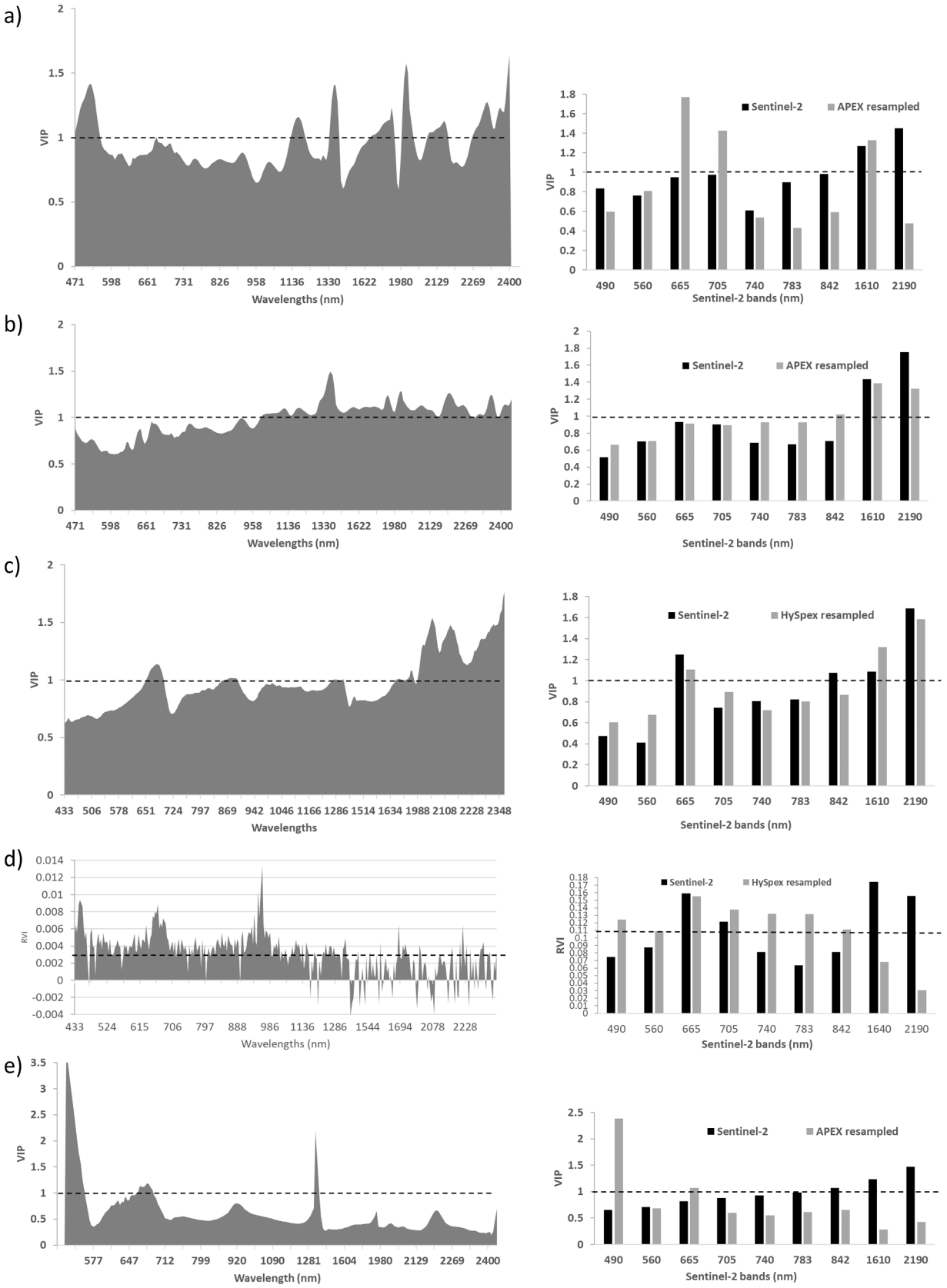


Fig. 5. Variance importance in projection (VIP) obtained by partial least square regression (PLSR) models for soil organic carbon (SOC) estimation in Oesling (a), Gutland (b), Demmin_S (c), Demmin_C (d), and loam belt (e) dataset. Relative variable importance (RVI) relating to random forest models was calculated for Demmin_C dataset (d). All the variable importance computations were carried out for hyperspectral (left-hand panel) and multispectral (right-hand panel) data.

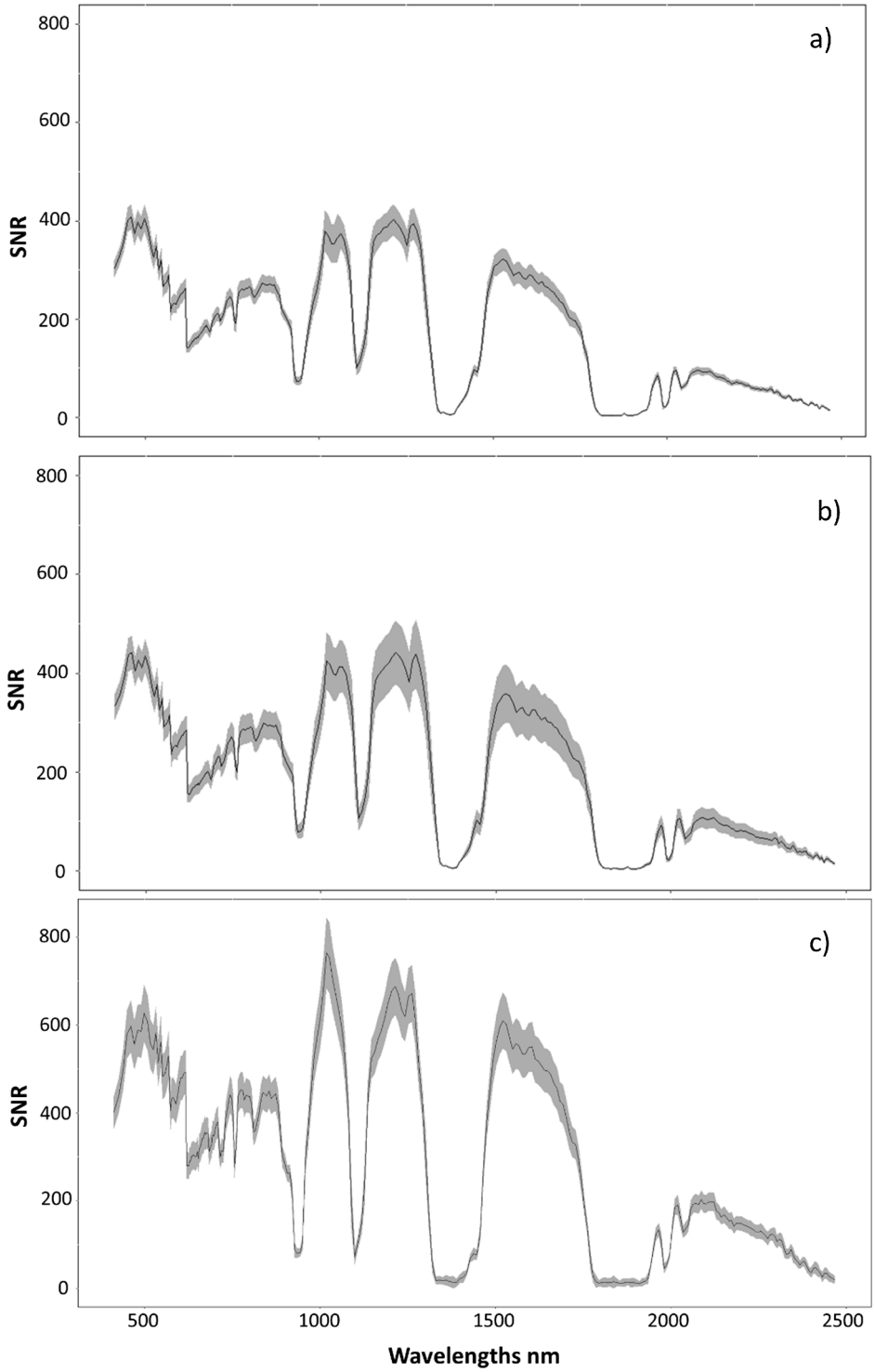


Fig. 6. Average Signal-to-noise ratio (SNR) of the APEX data associated with the location of the soil sample points collected within the Oesling (a), Gutland (b) and Loam belt area (c). The grey shadow area shows the standard deviation of the SNR values at each wavelength.

show an undulation which could be representative of a periodic phenomenon (cyclicity) called hole effect (Webster and Oliver, 2007). To model such a behaviour, its amplitude and frequency must be identified during the interpretation procedure (Pyrcz and Deutsch, 2003). It was not possible to define a periodic phenomenon but an explanation of such a behaviour could be found in the considerable differences in the parent material (underlying geological material) and elevation, which caused spatial variability of soil types in some parts. Such a problem could solved using polylines called faults, which allow taking into account the discontinuities between till soils and till soils with sandy layer (Fig. 8c) (see Table 5).

The two spherical structures of variogram for till soil with sandy layer showed two scales of variation of SOC (Fig. 11). Considering the large size of the field in this region, the first sill, which was reached at 2.25 km, could be related to the in-field variability. While the second sill at 6.9 km could be related to the between field variability due to different management practises. Concerning the till soil area, the sill was reached at a shorter distance (1 km) as compared with the sandy

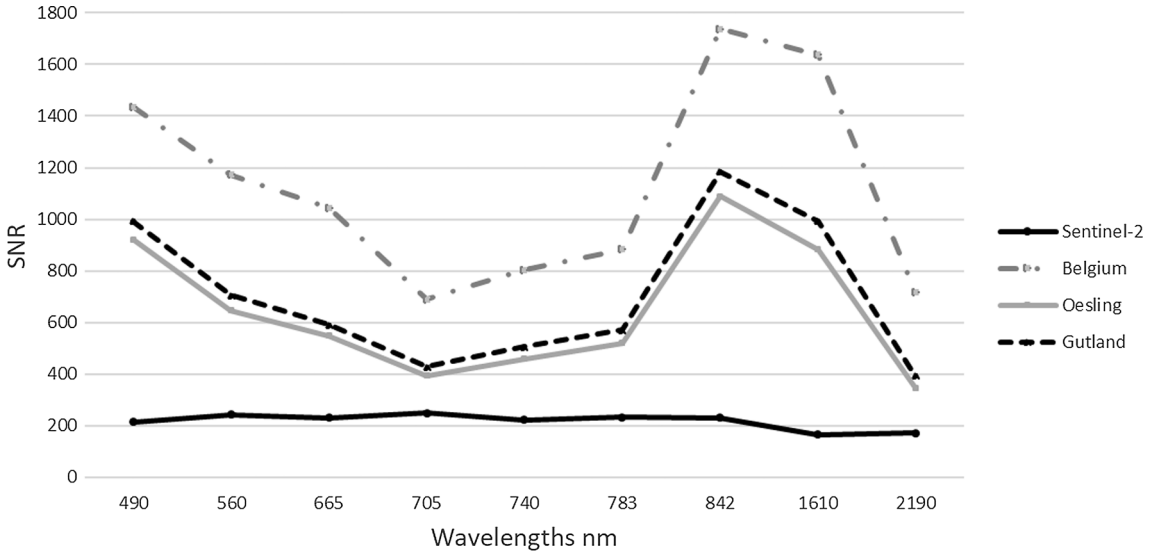


Fig. 7. Sentinel-2 signal-to-noise (SNR) ratio declared by European Space Agency and airborne SNR at each Sentinel-2 band averaged for the locations of the soil sample points in Oesling, Gutland and Loam belt area.

region, which is probably due to the large variability at short distance caused by soil and landscape discontinuities (kettle hole, grove, and peat area).

4. Discussion

Until now, multispectral satellite data were mainly used for soil classification as support for digital soil mapping (McBratney et al., 2003; Mulder et al., 2011), but the spatial and spectral resolution of the recent multispectral satellite sensors allows progress beyond qualitative soil assessments, using multivariate statistics and ground chemical data. In particular, the new generation of multispectral imagers are equipped to acquire some wavelengths in the SWIR region, which includes absorption features related to SOC and soil texture. A real comparison between multispectral (satellite) and hyperspectral (satellite or airborne) data would clarify the influence of spectral and spatial resolution on soil property estimation, providing clear information on the actual capability of the new generation of satellite sensors. Both

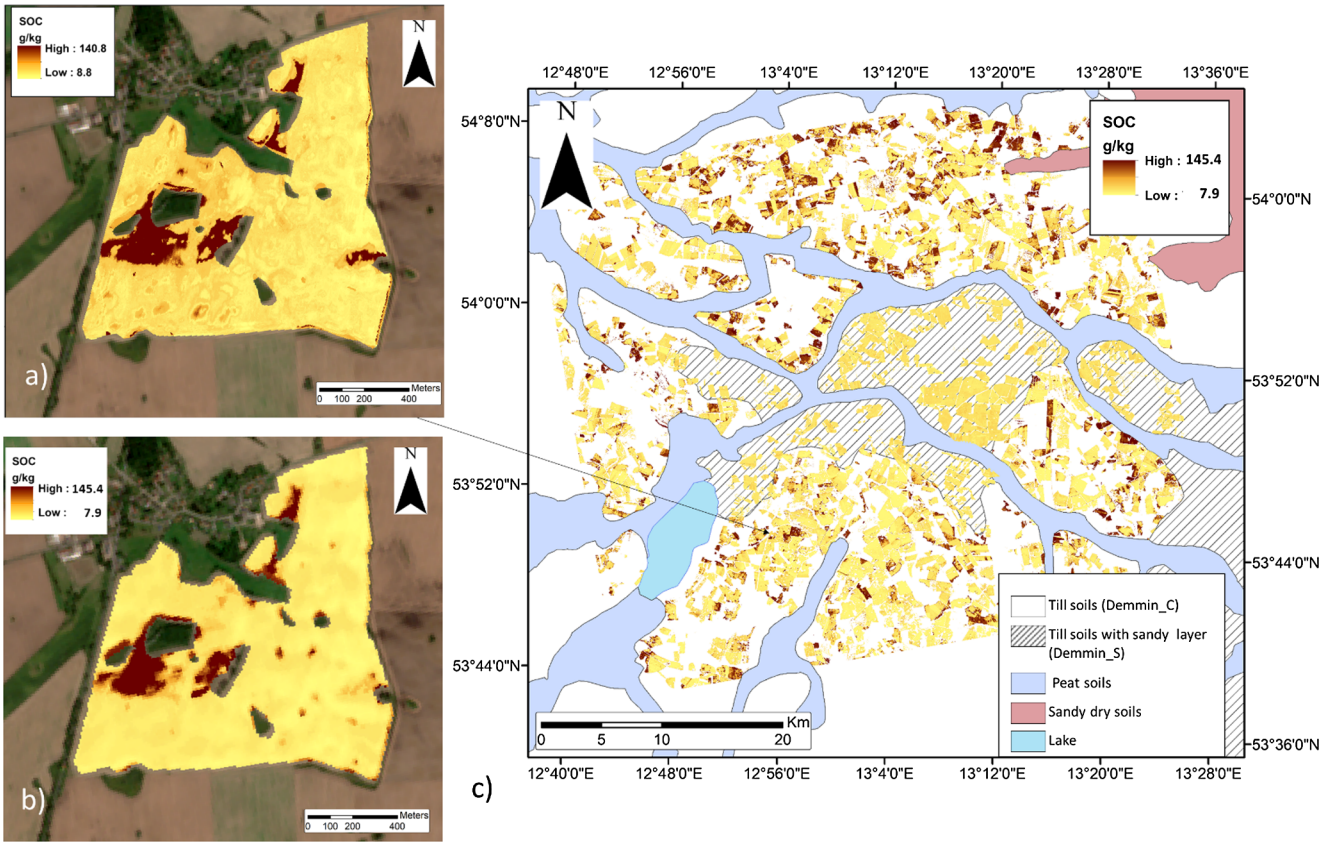


Fig. 8. Soil organic carbon (SOC) maps of a singular field in Demmin_C area obtained by HySpex (a) and real Sentinel-2 (b) data. On the right hand side (c), the SOC map at regional scale retrieved from Sentinel 2 data in the Demmin_C and Demmin_S sub-regions.

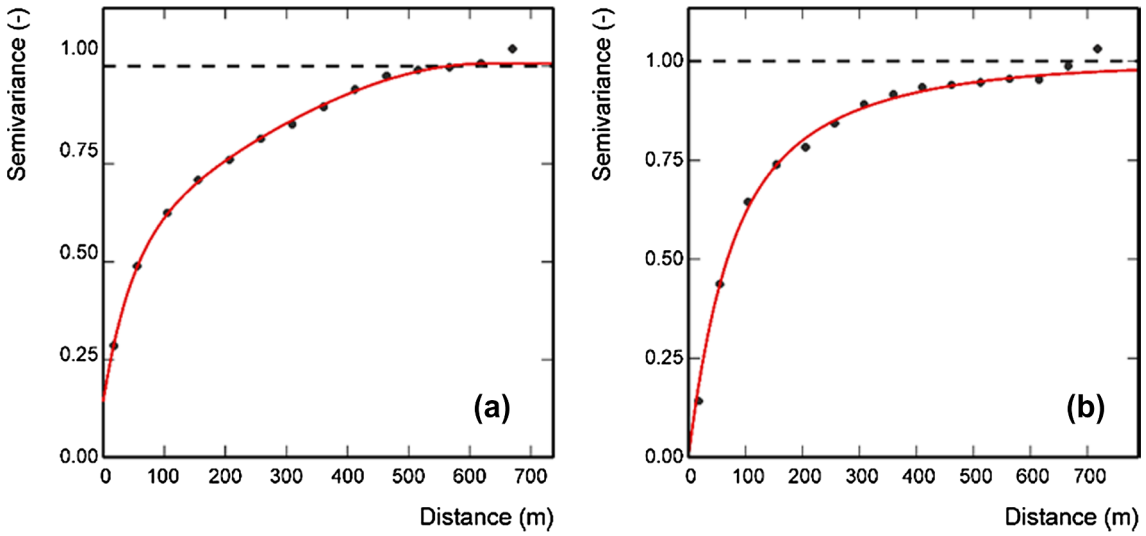


Fig. 9. Variograms of the Gaussian HySpex (a) and Sentinel-2 (b) SOC data at field scale. The filled points are the experimental semivariance values whereas the solid lines are the model of variograms. The horizontal dashed lines indicates the experimental variance.

Table 4
Variogram parameters of the fitted model for Gaussian SOC data of field HySpex and Sentinel-.

Variable	Model	Sill (–)	Range (m)
SOC field HySpex	Nugget effect	0.1423	
	Exponential*	0.4144	431.73
	Spherical	0.4506	628.68
SOC field Sentinel-2	Exponential*	0.5548	526.83
	Exponential*	0.4338	1924.73

* The exponential model has no finite range and here is reported the practical range. It is the distance at which the variogram value equals 95% of the sill variance (Webster and Oliver, 2007).

Table 5
Variogram parameters of the fitted model for Gaussian SOC data of regional Sentinel-2 for till soil and till soil with sandy layer using faults.

Variable	Model	Sill (–)	Range (m)
SOC Regional Sentinel-2 Till soil	Nugget effect	0.3718	
	Exponential*	0.6538	1035.66
SOC Regional Sentinel-2 Till soil with sandy layer	Nugget effect	0.5511	–
	Spherical	0.3870	2249.31
	Spherical	0.0588	6928.12

* The exponential model has no finite range and here is reported the practical range. It is the distance at which the variogram value equals 95% of the sill variance (Webster and Oliver, 2007).

Castaldi et al. (2016) and Gomez et al. (2018) explored the capability of the spectral resolution of S2 data for soil properties prediction, but neither of them used actual images nor did they take the spatial resolution into account. Castaldi et al. (2016) highlighted the advantage of using simulated hyperspectral satellite data instead of S2 for both clay and SOC prediction, however the prediction accuracy obtained with S2 was equal to that achieved with simulated Hyperion data, which had a very low SNR ratio in the SWIR. Also Stevens et al. (2010)

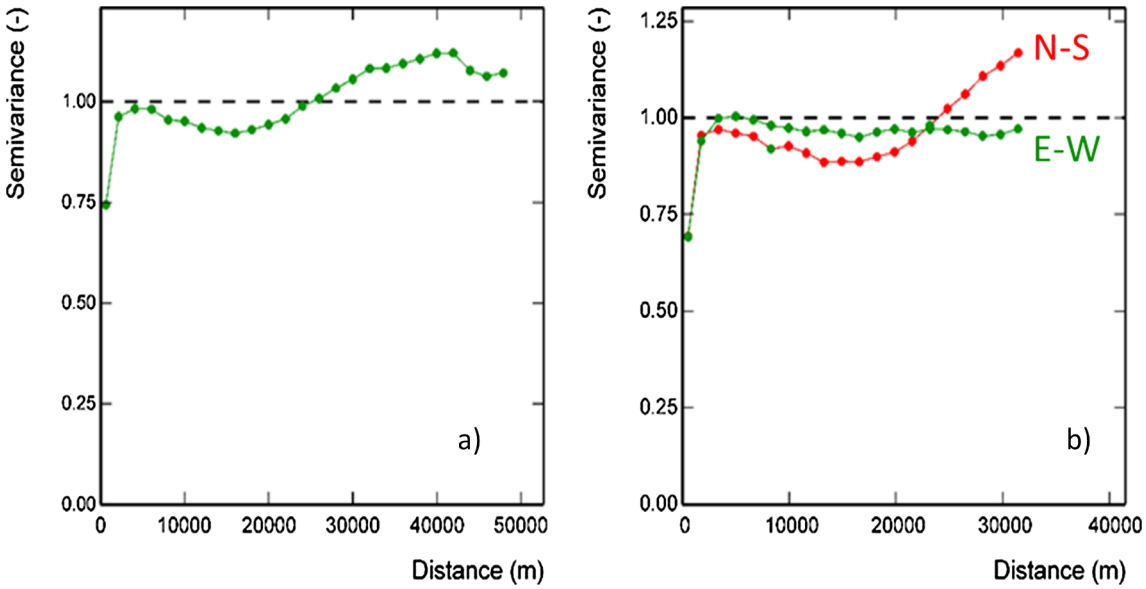


Fig. 10. Experimental variograms of SOC at regional scale computed: omnidirectional (a), along north-south (N-S) and east-west (E-W) directions (b). The filled points are the experimental semivariance values. The dashed lines are the experimental variances.

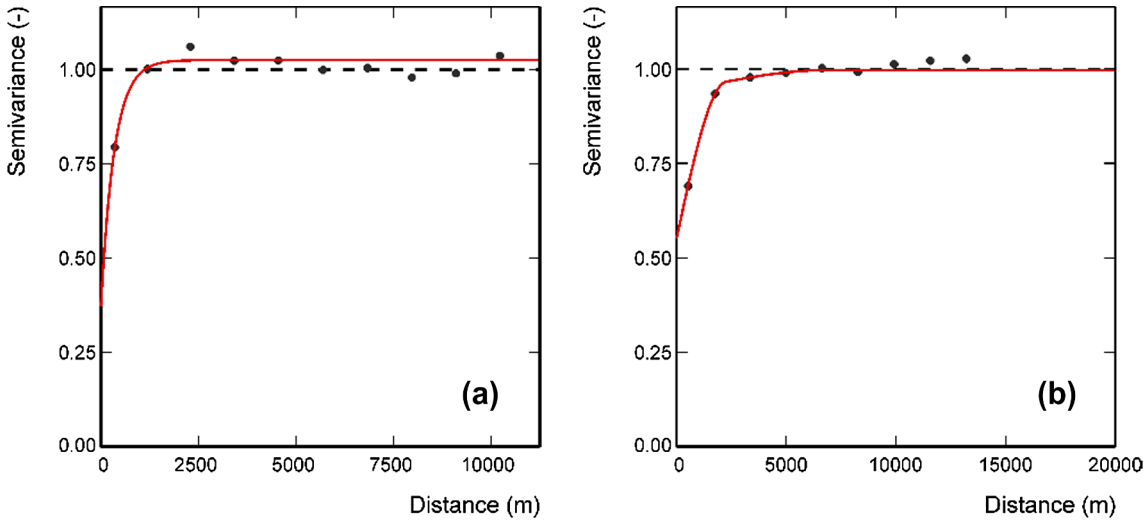


Fig. 11. Experimental variograms of SOC calculated for till soils (a) and till soils with sandy layer (b) and using faults. The filled points are the experimental semivariance values whereas the solid line is the fitted model of variogram. The dashed lines are the experimental variances.

highlighted the importance of the SNR for SOC prediction in Luxembourg using the Airborne Hyperspectral Sensor 160 (AHS), since they measured very low SNR values beyond 2000 nm (< 20), they decided to remove part of the SWIR bands before the statistical analysis for SOC prediction. For the Belgium and Luxembourg flight campaigns, we measured a better SNR within the SWIR region as compared to that reported by Stevens et al. (2010). The reduced energy collected by the hyperspectral sensor in narrow spectral bands and the low solar irradiance entails a lower SNR in SWIR region of hyperspectral sensors than for multispectral ones, which have broad bands (Castaldi et al., 2016). The signal quality in the SWIR region can strongly affect the SOC prediction accuracy. This is due to the main compounds of the organic matter that affect reflectance around 2100 nm (cellulose) and 2300 nm (lignin and starch) (Ben-Dor et al., 1997). The analysis of the S2 data showed that the two SWIR broad bands centred at 1610 and 2190 nm are sufficient to satisfactorily predict SOC content even if their VIP values depend on the differences between the other spectral regions in terms of SNR (Fig. 5). The width of the two S2 bands in the SWIR is probably irrelevant, because each of the main components of the organic matter (lignin, cellulose, chlorophyll, etc.) affects the spectrum at different wavelengths (Ben-Dor et al., 1997), and most of their spectral features fall within the spectral range of the B11 (1540–1680 nm) and B12 (2070–2310 nm). Also Gholizadeh et al. (2018) obtained satisfactory results for SOC prediction using S2 data in four agricultural sites in Czech Republic (RPD = 1.26–2.05). The SOC estimation accuracy obtained here are slightly higher than those previously obtained in Luxembourg by Castaldi et al. (2018b) (RPD = 1.7) and in Demmin region by Gerighausen et al. (2012) (RPD = 1.8), both using hyperspectral VNIR-SWIR airborne data. In particular, a strong match between the VIP values computed by Gerighausen et al. (2012) and ours exists: the highest VIP values were found in the visible between 650 and 700 nm and in the SWIR region after 2000 nm for both (Fig. 5c and d). On the other hand, the five investigated areas showed contrasting VIP values, for example the VIP values at the beginning of the VIS region (450–560 nm) are higher than 1 in Oesling and Loam belt, while they are lower than 1 in Gutland and Demmin_S. This inconsistency is partly due to the heterogeneity of the organic matter composition, which entails a large range of spectral response. Although a direct relation between darkness of the soil and quantity of organic matter exists, the strength of this relation is not the same everywhere and it can be influenced by the parent material and soil moisture content (Castaldi et al., 2018a). Actually, the regions where the VIS wavelengths are more important for SOC prediction are those where the soil is generally darker (Demmin_C, Loam belt and Oesling). Moreover, this lack of overlap between VIP plots confirms the difficulties when extrapolating empirical models in areas with different soil conditions.

The SNR and the spectral resolution in the SWIR region is probably even more crucial for clay prediction than for SOC, because all the most important spectral features related to clay minerals are located beyond 2100 nm: around 2170 nm (kaolinite), 2200 nm (illite, montmorillonite, and kaolinite) and 2360 nm (illite). Consequently, for the clay content retrieval, the bandwidth of B12 is probably not useful to exploit the narrow spectral features related to clay minerals. Gholizadeh et al. (2018) observed a decrease in clay prediction accuracies moving from hyperspectral airborne to S2 data.

The resampling of airborne data according to the S2 bands allowed investigating the importance of the spectral resolution and SNR, keeping constant the spatial resolution and the calibration dataset. The comparison between multispectral (S2) and hyperspectral (APEX or HySpex) did not show remarkable differences in terms of SOC prediction accuracy (Table 3). The resampling procedure entailed on one hand the decrease of spectral resolution and on the other hand the increase of the signal due to the high number of photons of collected with broader bands (Seidel et al., 2018). Thus, the loss of spectral information was probably balanced by the improvement of the SNR. Generally, VIP values of SWIR bands decreased to the benefit of red and NIR bands (Fig. 5) for resampled data. These differences of VIP values can be explained by the loss of spectral resolution as compared to airborne hyperspectral data. After all, the differences of SNR between SWIR and VNIR bands in the S2 spectra are clearly smaller than those of the resampled APEX data (Fig. 7). The increase of SNR differences between the band centred at 2190 nm and VNIR and the reduced spectral resolution of the resampled data entailed a decreasing of the VIP values of the SWIR band; consequently, the prediction models mainly exploited the VNIR region, where other absorption features related to organic matter are located (Ben-Dor et al., 1997). The wavelengths of the organic matter spectral features in the VIS are very close to the central wavelengths of the S2 bands, especially the red band. This proximity entailed a VIP value higher than 1 for the S2 band centred at 665 nm in Demmin_S and Demmin_C and very close to 1 in the two Luxemburg regions. On the other hand, the VIP values of the S2 bands centred at 665 and 705 nm is considerably lower than those provided by APEX or resampled data for the Oesling dataset (Fig. 5a). As explained previously, S2 and airborne images were acquired at different dates and the bare soil conditions were visually checked only during the airborne flight campaigns. The Luxembourg S2 image was acquired in May,

when probably maize was emerging in some fields that were classified as 'bare soil' (NDVI < 0.25) and consequently, the lowering of VIP of the red and red edge bands is probably due to the influence of the maize seedlings on these wavelengths highly sensitive to vegetation. In addition, the presence of crop residues and weeds in some fields in the loam belt at S2 acquisition (end of August) could be the cause of the strong VIP differences between actual S2 and airborne data at 490 nm, a wavelength close to the chlorophyll *a* and *b* absorbance peaks. A consistent presence of green or dry vegetation (> 20%) can strongly affect the shape of spectral reflectance (Bartholomeus et al., 2011), and consequently, it can influence the prediction accuracy of the soil properties. Although, hyperspectral imagery has proven to be able to separate pure soils from soils with crop residues (i.e. non-photosynthetic vegetation; e.g. Bartholomeus et al., 2011; Bayer et al., 2016), the capability of the multispectral S2 data for spectral unmixing techniques still needs to be tested.

In addition to the sensor quality, the variability of the calibration dataset is a crucial factor for the soil property prediction accuracy by regression models (Ben-Dor et al., 2002). Indeed, the German and Luxembourg datasets, which have a large variability in terms of SOC content, provided high RPD values. While for the Belgian dataset, where the standard deviation is very low and the range is narrow (Table 2), the RPD never exceeded 1.3. These results highlight the importance to adopt a suitable sampling strategy to collect a calibration dataset representative of the investigated area and at the same time including, as much as possible the full range of SOC values.

In the light of the satisfactory SOC prediction accuracy achieved by the S2 data, we investigated the capability of the S2 to explain the infield and regional SOC variability. Although Gomez et al. (2015) indicated that a spatial resolution of 10 m could be sufficient to detect the spatial variability of most of soil properties, the geomorphological characteristics in the Demmin area cause some abrupt changes of SOC content at short distances. These sudden changes of SOC content are particularly frequent in the Demmin_C sub-region, where the soils with moderate SOC content surround some small areas with very high SOC content characterized by kettle holes where water logging is more frequent, influencing both biomass production and reducing decomposition (Blasch et al., 2015). Some small and large kettle holes areas, a common feature for ground moraine landscapes, can be detected (Fig. 8). Both airborne (Fig. 8a) and S2 (Fig. 8b) data described the same SOC patterns within the field, thus the reduction of spatial resolution from 4 to 10 m appears not to entail a loss of detail in the SOC map, also considering that the spatial resolution of the SWIR bands provided by the S2 sensor is 20 m. Both SOC maps detected a large area on the western side characterized by moderate and very high SOC content, while on the eastern side the SOC content is generally moderate, apart from some elliptical areas with high SOC values related to small kettle holes. The behaviour of two variograms (Fig. 9) is in agreement with what one theoretically would be expected to the increase of the pixel size (support), i.e. the range of variogram increase from HySpex to Sentinel2 SOC whereas the nugget effect decreases (Table 4). In the light of the goodness of the S2 maps obtained in Demmin, we can assume that the spatial and spectral resolution of the S2 sensor are able to describe the SOC variability at short range in the other study areas, where the infield variability is considerably lower than in Demmin.

The short revisit time of the S2 constellation combined with the Sentinel hub web service provides a large amount of freely available satellite data. Thus, we are witnessing an improved access to satellite data, which will attract new users towards remote sensing. Moreover, the availability of many acquisition dates during the year increases the possibility to detect a large number of bare soil fields observed under cloud-free conditions and thus to extend the SOC maps by mosaicking the output derived from different images. Furthermore, this large amount of data could help to develop specific algorithms for detecting the conditions under which the soil prediction models can be applied. In particular, the multi-temporal series could be employed to detect croplands (Galford et al., 2008; Rogge et al., 2018) or to reduce the influence of soil moisture (Castaldi et al., 2015) and crop residues on the prediction accuracy.

5. Conclusions

We investigated the capability of Sentinel 2 bare soil images to predict the organic carbon content of the topsoil in croplands based on PLSR and RF models and ground truth model calibration. We compared the S2 results with those obtained using airborne hyperspectral data in Luxembourg, Belgium and Germany. The prediction accuracy obtained by S2 data is very similar to that retrieved by airborne hyperspectral data. After the spectral resampling of the airborne data according to S2 band, the prediction accuracy did not change for four out of five subregions. However, the importance of SWIR bands decreased and these differences of VIP values could be explained by the loss of spectral resolution as compared to hyperspectral data and the strong difference in terms of SNR between SWIR and other spectral regions. The investigation of the spatial variability of the SOC maps derived by S2 data has shown no substantial differences with the SOC maps obtained by hyperspectral airborne data. Thus the spatial resolution and spectral characteristics of the S2 are adequate to describe SOC variability both within field and at regional scale. The level of details of and the extent of the area covered by these maps make them suitable both for environmental monitoring related to global warming for which a coarse spatial resolution over large areas is usually sufficient and for precision agriculture, where a finer spatial resolution at the field scale is required in order to set specific site management. In the light of these promising results, further research will need to test the capability of S2 data for SOC or other soil properties prediction in other regions, where appropriate, using covariates to increase the prediction accuracy. The development of new satellite sensors, having a higher spectral resolution in the SWIR region than that provided by S2 sensors, could be an advantage especially for soil properties characterized by narrow spectral features, as clay and calcium carbonate content.

Acknowledgements

The research was funded by the Belgian Federal Science Policy Office (BELSPO) as part of the PROSOIL project "The evaluation of forthcoming satellites for mapping topsoil organic carbon in croplands" (Contract SR/10/327). We are grateful to Marco Bravin of the Earth and Life Institute of the Université Catholique de Louvain (UCLouvain) for essential organic carbon measurements.

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